

Origins and applications of the Montroll-Weiss continuous time random walk[★]

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Abstract. The Continuous Time Random Walk (CTRW) was introduced by Montroll and Weiss in 1965 in a purely mathematical paper. Its antecedents and later applications beginning in 1973 are discussed, especially for the case of fractal time where the mean waiting time between jumps is infinite.

1 Background for the CTRW

The origin of probability dates back to determining odds in games of chance. The first classic book on probability was Jacob Bernoulli's *Ars Conjectandi* (The Art of Conjecturing) published posthumously in 1713. If a player could win or lose a coin in a round of play, then his fortune would go up or down by one coin. This is a random walk on a lattice of spacing one, but with absorbing boundaries at the point where a player loses his last coin or wins all the coins. If there are two players and one player has m coins, then on the next round of play he will have either $m + 1$ or $m - 1$ coins. This is an example of a Markov chain where the next number of a player's coins depends on his current number and not on earlier outcomes. Markov [1] introduced his chains in a desire to study the probabilities of dependent events. As an example, he picked a Russian novel and in his 1913 work calculated the probability that a letter chosen at random would be a consonant with probability C or a vowel with probability V . If letters appeared independently at random then the probability of two vowels in a row would be V^2 , but this result was not found. Markov created a 4-state system with a V or C followed by a V or C (V and C corresponding here to letter types) with the result that a V followed by a C was more likely than a V followed by a V, so the next state of the system depended on the present state. What if there was a time lag for making transitions among the Markovian states. That is precisely the point addressed by Paul Lévy [2] in 1954 at the inaugural International Congress of Mathematicians in his article "Processus Semi-Markoviens". Unlike the early gambling questions, Lévy was concerned with theorems about limit distributions. He wrote, "Chaque fois que le système

étudié arrive à un état A_n , la durée U_n de son séjour dans cet état est une variable aléatoire non négative", my translation: each time the system under study arrives at a state A_n the duration U_n of its sojourn in that state is a non-negative random variable. With Lévy's paper, waiting times in Markov processes made their debut and the process was dubbed semi-Markovian. Lévy's transitions between states were given in terms of rates while Markov used probabilities. A fixed rate implies an exponential distribution of waiting times. Fixed rate waiting times were introduced again, independently, in 1955 by W.L. Smith [3] in his article in the Proceedings of the Royal Society "Regenerative Stochastic Processes". Waiting times were to be introduced again in an elegant paper by Elliott Montroll and George Weiss [4] in 1965 for random walks on a lattice and in the 1970's taken to the realm of infinite mean waiting times by Harvey Scher and collaborators [5–9]. It was in the Scher-Lax [5] paper that the Montroll-Weiss process was first labeled as CTRW. These works [5–9] inaugurated the field of anomalous diffusion for transport.

Elliott Montroll was a renowned physicist with classic works on a variety of topics including works on a number of lattice problems. These include lattice vibrations, the Ising model and random walks. Random walks were an old topic that seemed fully understood and explored, belonging to textbooks and not having novel research directions. It must have seemed odd to my fellow graduate students that my thesis was on random walks, while they investigated modern topics, but my work was eventually the cover article of *Physics Today* [13]. Montroll was well known for his exceptionally well written papers to explain and educate the reader. The Montroll-Weiss paper basically laid dormant until Harvey Scher applied it to the case of infinite mean waiting times which ignited a resurgence of interest in random walks. Benoit Mandelbrot's work on fractals furthered the interest and application of random walks with a fractal nature. George Weiss was unusually prolific and contributed many works on random walks.

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I remember one time walking into his office, jointly tackling a problem on the blackboard of the mean number of times a random walker visits a line on a three dimensional lattice, and walking out an hour later with a joint publication. After the Montroll-Weiss CTRW paper, he made many more contributions to pure and applied random walks [14] including a CTRW analysis of financial markets [15].

Elliott Montroll mentioned in his lectures that he greatly admired the works of Lord Rayleigh. Rayleigh is well known for his theory of sound, winning a Nobel Prize for the discovery of argon, and he also wrote about random walks. Rayleigh's experimental work succeeded in separating argon from air through a cascade series of filters. Inspired by Rayleigh's work, Montroll, working for the Kellogg Corporation in the early 1940's, mathematically modeled the gaseous diffusion process for the separation of uranium isotopes at Oak Ridge as a random walk through a cascade of membranes. This led to Montroll's first random walk paper [10] published in 1946 entitled "A Note on Bessel Functions of Purely Imaginary Argument" which did not mention, but was related to the uranium isotope separation project. In 1964 Montroll wrote another random walk paper [11] "Random Walks on Lattices" for a proceedings and then together with his former graduate student, George Weiss [4] they wrote in 1965 the famous paper "Random Walks on Lattices II" in the Journal of Mathematical Physics (where Montroll was the editor). Their approach was the use of lattice Green's functions and then Tauberian theorems for asymptotic analysis. In the 1964 random walk paper Montroll cites the famous 1928 mathematics paper [12] by Richard Courant, Kurt Friedericks and Hans Lewy (not Paul Lévy) for a Green's function (in the German, Greenschen Formeln) approach for numerical computation for solving partial differential equations on grids including the diffusion equation. Why was Courant interested in numerical analysis? For one, the Courant and Hilbert book "Methods of Mathematical Physics" was published in 1924 and the second volume was devoted to partial differential equations. A second connection might be that Courant was married to Carl Runge's daughter Nina and it was Carl Runge who pioneered numerical methods in physics. His 1900 work (followed by Martin Kutta in 1901) introduced a method to numerically iteratively integrate differential equations. Back then physics was considered as an experimental science. Being employed as a theoretician was unusual (Einstein helped start the notion of a solely theoretical physics career), but making a living for numerical work was even more difficult, but Runge's work has been a lasting contribution. The Courant-Friedericks-Lewy paper is famous in the mathematics literature and it influenced Montroll, perhaps because Montroll spent some time at the Courant Institute.

2 The CTRW lattice Green's function

The Montroll-Weiss paper was a masterful calculation of evermore complicated random walk quantities, such as first passage time moments, the number of points visited

r times in an n -step walk, and exact probabilities for return to the origin for a variety of 3D lattices (while Feller's book still used approximate values, despite Montroll saying that he communicated the exact results to Feller). To start the CTRW discussion, let's review this Green's function method.

Consider a random walk on an infinite periodic lattice with unit spacing. Let $P_n(l)$ be the probability that the random walker is at site l on the n th step of the walker and we assume the walk began at the origin, so $P_0(l) = \delta_{l,0}$. Let $p(l)$ be the probability that a jump is of a vector l displacement, then

$$P_{n+1}(l) = \sum_{l'=-\infty}^{\infty} P_n(l-l')p(l'). \quad (1)$$

The random walk generating function $G(l; z)$ is defined as

$$G(l; z) = \sum_{n=0}^{\infty} P_n(l) z^n. \quad (2)$$

Multiplying equation (1) by z^{n+1} and summing over all n and remembering the initial condition, the generating function takes the form of a Green's function,

$$G(l; z) - z \sum_{l'=-\infty}^{\infty} G(l-l'; z) p(l') = \delta_{l,0}. \quad (3)$$

Fourier transforming from l to k we arrive at,

$$G(k, z) = \frac{1}{1 - zp(k)}. \quad (4)$$

Now to add the time variable to derive the CTRW. It is only towards the end of the Montroll-Weiss paper, in Section 5 that the CTRW is introduced. While the formalism was elegant, the main asymptotic result was that n , the number of jumps, was replaced by $t/\langle t \rangle$ with $\langle t \rangle$ being the mean waiting time between jumps and t is the time of the ongoing process. The work attracted no attention until the Scher and Lax work, on ac impurity conduction in an amorphous semiconductor, considered a hopping process where the mean waiting time for a jump was non-exponential.

The CTRW begins with $Q(lt)$ the probability density to reach site l exactly at time t , and $\psi(t)$ the probability per unit time that the walker waits a time t between jumps. The equation for $Q(lt)$ is given by,

$$Q(l, t) = \sum_{l'=-\infty}^{\infty} \int_0^t Q(l-l', t-\tau) p(l') \psi(\tau) d\tau + \delta_{l,0} \delta(\tau). \quad (5)$$

Fourier transforming to k -space and Laplace transforming to s -space yields

$$Q(k, s) = \frac{1}{1 - \psi(s)p(k)} \quad (6)$$

which equates to the Green's function $G(k; z)$ where the dummy variable z has been replaced by $\psi(s)$. The probability $P(l, t)$ occurs when the walker reaches site l at time

$t - \tau$ and then does not jump for at least a time τ . This gives the famous Montroll-Weiss formula

$$P(l, t) = \int_0^t Q(l, t - \tau) \Phi(\tau) d\tau$$

with $\Phi(\tau) = \int_\tau^\infty \psi(t') dt'$ the probability that the next jump takes place after time τ .

In Fourier-Laplace space we have the most used form for the CTRW

$$P(k, s) = \frac{1}{1 - \psi(s)p(k)} \frac{1 - \psi(s)}{s}. \quad (7)$$

3 A master equation for the CTRW

Harvey Scher was working at the Xerox Research Center in Webster, NY, close by to the U. of Rochester where Elliott Montroll was a professor and a consultant to Xerox. Scher was able to interpret some unusual results for charge transport in amorphous semiconductor films used in xerography in term of the Montroll-Weiss random walk, but with a waiting time density with a long algebraic tail. Montroll was interested in expressing the random walk as a master equation that would incorporate the case of waiting times densities with an infinite first moment. It was obvious that using a constant rate approach would not work and that one needed to incorporate a memory. As his graduate student, Montroll suggested this problem to me which resulted in a 1973 paper. The following [7] memory master equation for the CTRW is derived as follows,

$$\begin{aligned} \frac{\partial P(l, t)}{\partial t} = & - \int_0^t P(l, t - \tau) \varphi(\tau) d\tau \\ & + \sum_{l'} \int_0^t P(l - l', t - \tau) p(l') \varphi(\tau) d\tau. \end{aligned} \quad (8)$$

$\varphi(t)$ has units of $1/t^2$ and it is related to the probability waiting time density $\psi(t)$ by performing the Laplace (over time) and Fourier (over space) transform to give the master equation in the form

$$sP(k, s) - P(k, t=0) = \varphi(s)p(k)P(k, s) - \varphi(s)p(k).$$

Solving for the transformed probability gives

$$P(k, s) = \frac{1}{s + \varphi(s) - \varphi(s)p(k)}.$$

If we set,

$$\varphi(s) = \frac{s\psi(s)}{1 - \psi(s)} \quad (9)$$

then

$$\begin{aligned} P(k, s) &= \frac{1}{s + \frac{s\psi(s)}{1 - \psi(s)} - \frac{s\psi(s)}{1 - \psi(s)}p(k)} \\ &= \frac{1}{1 - \psi(s)p(k)} \frac{1 - \psi(s)}{s} \end{aligned}$$

the CTRW result.

4 Condensed matter anomalous diffusion

Harvey Scher opened the anomalous diffusion era in 1973 in a work with Melvin Lax [5] on impurity band ac conductivity $\sigma(\omega)$ with a CTRW hopping model to explain the experimental behavior of Si of $\sigma(\omega) \sim \omega^{-8}$. Scher and Lax converted the cosine transformed velocity correlation function expression for ac diffusion $D(\omega)$ into the Fourier transform of the mean squared displacement to derive,

$$D(\omega) = -\frac{1}{6}\omega^2 \sum_r r^2 \int e^{i\omega t} P(r, t) dt \quad (10)$$

and $P(r, t)$ is calculated through the CTRW. The next step is to calculate $\psi(t)$.

If jump waiting times t grow exponentially with barrier height Δ as

$$t = t_0 e^{\frac{\Delta}{kT}} \quad (11)$$

then a distribution of Δ 's, $f(\Delta)$ induces a distribution of hopping times,

$$\psi(t) dt = f(\Delta) d\Delta.$$

If the barrier heights are distributed exponentially, using equation (11)

$$\begin{aligned} f(\Delta) &= e^{-\frac{\Delta}{\Delta_0}} \\ &= \left(\frac{t_0}{t}\right)^{\frac{kT}{\Delta_0}} \end{aligned}$$

yielding an algebraic asymptotic behavior

$$\psi(t) \sim t^{-1 - \frac{kT}{\Delta_0}} \quad (12)$$

which has an infinite first moment when $kT < \Delta_0$. If the jump waiting time increased exponentially with distance, $t = t_0 e^{\frac{R}{R_0}}$ and R was distributed in a Poisson manner of $e^{-\frac{R}{R_0}}$, then again $\psi(t)$ would have an asymptotic algebraic form.

Scher and Lax [5] derived a CTRW expression for the ac diffusion constant (in their notation)

$$D(\omega) = -\frac{i\omega}{6}\sigma^2 \frac{\tilde{\psi}(i\omega)}{1 - \tilde{\psi}(i\omega)} \quad (13)$$

where σ^2 is the mean squared jump distance and the waiting time probability has been Laplace transformed. For an algebraic waiting time regime (but with an eventual exponential fall off) they find a frequency domain with $D(\omega) \sim \omega^s$, with $s < 1$, i.e. a non-integer exponent, and thus the field of anomalous diffusion was born.

Next, together with Montroll, Scher tackled the puzzling experimental finding of time of flight measurement of current in amorphous As_2Se_3 a semiconductor material used as a film to capture images in xerography. Here one needed to calculate the mean biased motion of a random walker (hole transport in this material). When the first moment of $\psi(t)$ is infinite this does not mean

that there is no jump of a random walker, but that there is a distribution of jump times with infinite mean. If one calculates for a random walk biased in one direction with mean jump distance $\bar{x}$, then from a CTRW calculation where $\mathcal{L}^{-1}$ is the inverse Laplace transform

$$\langle x(t) \rangle = \sum_x x P(x, t) = \mathcal{L}^{-1} \bar{x} \frac{\psi(s)}{s(1-\psi(s))}. \quad (14)$$

For $\psi(t) \sim t^{-1-\beta}$ with $\beta < 1$, then $\psi(s) \sim 1 - \text{const. } s^\beta$ and

$$\langle x(t) \rangle \sim \bar{x} t^\beta \quad (15)$$

so the mean current $I(t) = d\langle x(t) \rangle / dt \sim t^{-1+\beta}$, a classic result in anomalous diffusion. When a charge reaches L (the sample size) it gets absorbed at an opposite polarity electrode at the end of the sample. A first passage time calculation accounts for losing charge at the sample boundary and changing the current to $I(t) \sim t^{-1-\beta}$. The fractional exponent has consequences with dimensional analysis. Normally a charge in an electric field E would have mobility μ and an average velocity $v = \mu E$. The transit time T across the sample would involve $L = vT = \mu ET$ so $1/T \propto E/L$. For the anomalous transport case with $\bar{x} \propto E$, then from equation (15) with $\langle x(t) \rangle = L$ and transit time T , $1/T \propto (E/L)^{1/\beta}$ resulting in the mobility having an E and L dependence. The fractional exponent forces the changing of dimensions, so now mobility depends on sample size. As there is a distribution of waiting times, the larger a sample the more likely a random walker will encounter sites (traps) with longer waiting times, so a size dependence for mobility occurs. This is an example of a fractal time process. I was lucky enough, as Elliott Montroll's graduate student to be introduced to this problem by Harvey Scher and have the opportunity to make a contribution [8] to the asymptotic analysis of this topic.

5 Stretch exponential decay from a many walker CTRW

Without going into the steps to calculate an expression for a first passage time within the CTRW formalism, we just note that the distinct sites visited by a random walk, $S(t)$, in 3D is asymptotically proportional to $t/\langle t \rangle$ when the first moment of $\psi(t)$ is finite, but $S(t) \sim t^\beta$, for $\psi(t) \sim t^{-1-\beta}$ when $\beta < 1$. If there is a reaction site at the origin and a concentration c of non-interacting random walkers not at the origin, the lifetime distribution $\Phi(t)$, that the origin has not been visited by time t is [9]

$$\Phi(t) = e^{-cS(t)}. \quad (16)$$

It is possible to use $S(t)$ because if a walker starting from the origin can reach $S(t)$ distinct sites in a time t , then walkers starting at these sites can reach the origin in a time t . The CTRW calculation derives the stretch exponential as a probability limit distribution when the mean waiting time between jumps is infinite.

6 A coupled space-time memory CTRW for supradiffusion

As a last topic, we consider a coupled space-time memory CTRW to derive a supra-diffusion case. Start with the CTRW for a coupled memory random walk, where

$$\Phi(x, t) = \psi(t|x) p(x) \quad (17)$$

where $\psi(t|x)$ is the probability per unit time that a jump takes time t given that it goes a distance x , and $p(x)$ is the probability that the jump is of length x . The probability density for a jump of duration t is $\psi(t)$. We choose for asymptotic long times that $\psi(t) \sim t^{-1-\alpha}$, $\alpha < 1$, so the first moment of $\psi(t)$ is infinite. Note that here $\psi(t)$ is a flight time, not a waiting time between jumps. Nevertheless, we'll use the CTRW formalism to see what that calculates.

For a jump of length x , we consider the case that the walker is always moving in a straight line of length x with a velocity $v(x)$ and at the end of the jump the walker reaches a turning point and continues with a new direction, jump length and velocity. We will calculate the mean square displacement.

Let's start by checking that, $\int \Phi(x, t) dx = \psi(t)$. In our specific case, let

$$\psi(t|x) = \delta\left(t - \frac{x}{V(x)}\right) = \delta(t - x^\beta) \quad (18)$$

that means that $V(x) = x^{1-\beta}$, $\beta = 1$ for ballistic motion, and $\beta < 1$ for supra-diffusive motion.

A distribution of flight duration times will induce a distribution of distances, i.e.

$$\psi(t) dt = p(x) dx.$$

For $t = x^\beta$, $\frac{dt}{dx} = \beta x^{\beta-1}$, and choosing $\psi(t) \propto t^{-1-\alpha}$, $\alpha < 1$, then

$$\begin{aligned} p(x) &= \beta t^{-1-\alpha} x^{\beta-1} \\ &= \beta x^{-1-\alpha\beta}. \end{aligned}$$

Let,

$$\begin{aligned} I &= \int \Phi(x, t) dx = \int \psi(t|x) p(x) dx \\ &= \int \delta(t - x^\beta) p(x) dx. \end{aligned}$$

Using

$$\int f(x) \delta(g(x)) dx = \frac{f(x_0)}{\left| \frac{dg(x_0)}{dx} \right|},$$

with $f(x) = p(x)$ and $g(x) = t - x^\beta$, $\frac{dg}{dx} = -\beta x^{\beta-1}$, and $x_0 = t^{\frac{1}{\beta}}$ being the zero of $g(x)$, then,

$$\begin{aligned} I &= \frac{p\left(t^{\frac{1}{\beta}}\right)}{\beta x_0^{\beta-1}} = \frac{\beta t^{\frac{1}{\beta}(-1-\alpha\beta)}}{\beta t^{\frac{1}{\beta}(\beta-1)}} \\ &= t^{-1-\alpha} \\ &= \psi(t). \end{aligned}$$

Now we calculate the mean squared displacement

$$\begin{aligned}\langle x^2(t) \rangle &= \int x^2 P(x, t) dx \\ &= - \lim_{k \rightarrow 0} \frac{\partial^2}{\partial k^2} \int e^{ikx} P(x, t) dx \\ &= - \lim_{k \rightarrow 0} \mathcal{L}^{-1} \frac{\partial^2}{\partial k^2} \int e^{ikx} e^{-st} P(x, t) dx dt \quad (19)\end{aligned}$$

where $\mathcal{L}$ is the Laplace transform. In Laplace-Fourier space, using the CTRW equation the mean squared displacement (1D case, no bias), is

$$\begin{aligned}\lim_{k \rightarrow 0} \langle x^2(k, s) \rangle &= - \lim_{k \rightarrow 0} \frac{1 - \psi(s)}{s} \nabla^2 \frac{1}{1 - \Phi(k, s)} \\ &= \lim_{k \rightarrow 0} - \frac{1 - \psi(s)}{s} \frac{\nabla^2 \Phi(k, s)}{[1 - \Phi(k, s)]^2}.\end{aligned}$$

Therefore,

$$\begin{aligned}\lim_{k \rightarrow 0} \langle x^2(k, s) \rangle &= - \frac{1}{s(1 - \psi(s))} \nabla^2 \Phi(k, s) \\ &= - \lim_{k \rightarrow 0} \frac{1}{s(1 - \psi(s))} \nabla^2 \int e^{-st} e^{ikx} \delta(t - x^\beta) \\ &\quad \times p(x) dx dt \\ &= \frac{1}{s(1 - \psi(s))} \int e^{-st} dt x^2 \delta(t - x^\beta) p(x) dx \\ &= \frac{1}{s(1 - \psi(s))} \int e^{-st} dt \left(t^{\frac{2}{\beta}} p\left(t^{\frac{1}{\beta}}\right) \right) / \beta t^{\frac{\beta-1}{\beta}}.\end{aligned}$$

For $\psi(t) \propto t^{-1-\alpha}$, then $p(x) \propto \beta x^{-1-\alpha\beta}$, so $p(t^{1/\beta}) \propto \beta t^{-\frac{1+\alpha\beta}{\beta}}$.

Yielding asymptotically,

$$\begin{aligned}\langle x^2(s) \rangle &= \frac{1}{s^{1+\alpha}} \int e^{-st} dt \left(t^{\frac{-1}{\beta}} t^{-\alpha} t^{2/\beta} t^{1/\beta} t^{-1} \right) / \beta \\ &\propto 1/s^{\frac{2}{\beta}+1}\end{aligned}$$

for $\beta = \frac{2}{3}$ (the Richardson turbulent diffusion case) $\langle x^2(s) \rangle \propto s^{-4}$ and $\langle x^2(t) \rangle \propto t^3$. For ballistic motion, $\beta = 1$ and $\langle x^2(t) \rangle \propto t^2$.

For a coupled memory random walk with continuous motion, turning points and no waiting times, this process would not be a CTRW process and different results are possible including a divergent second moment.

7 Final remarks

Elliott Montroll and his former student, George Weiss, both working nearby in Maryland produced a beautiful paper that allowed physicists to calculate detailed random

walk properties. The extension to the continuous time case was almost as an after-thought and not thoroughly explored in the paper. It was only when experiments at Xerox exhibited unusual behavior of mobility in amorphous semiconductors that the CTRW came to the fore, first with Scher and Lax and then with Scher and Montroll. Scher overlapped with Lax at Bell Labs and later with Montroll in the Rochester area. In the days before the internet proximity fostered joint work. Anomalous diffusion in condensed matter arose from Scher and colleagues' works daring to use in a sophisticated manner probability distributions with infinite moments in physics problems to explain experiments. The critical exponent in the waiting time distribution was temperature dependent and a phase from anomalous to normal transport occurred as the temperature was increased. Where the equilibrium phase transition theory encountered a diverging correlation length, and universal critical exponents, the anomalous diffusion theory for non-equilibrium transport encountered diverging waiting times, and non-universal transport exponents, and dynamical phase transitions as a function of temperature. The CTRW has now been generalized, spread to many scientific fields and has been the topic of conferences and many thousands of citations. At the 50 year point the Montroll-Weiss CTRW has turned out to create a family of colleagues and friends and that has spanned disciplines, continents and generations.

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